

DEVOPS PROGRAM 4:

Integrating Git with Docker for Source-Controlled Application Builds

PROJECT STRUCTURE:

- program4/
 - Dockerfile
 - index.js
 - package.json
 - package-lock.json

Step 1: Create Project Directory

`mkdir program4`

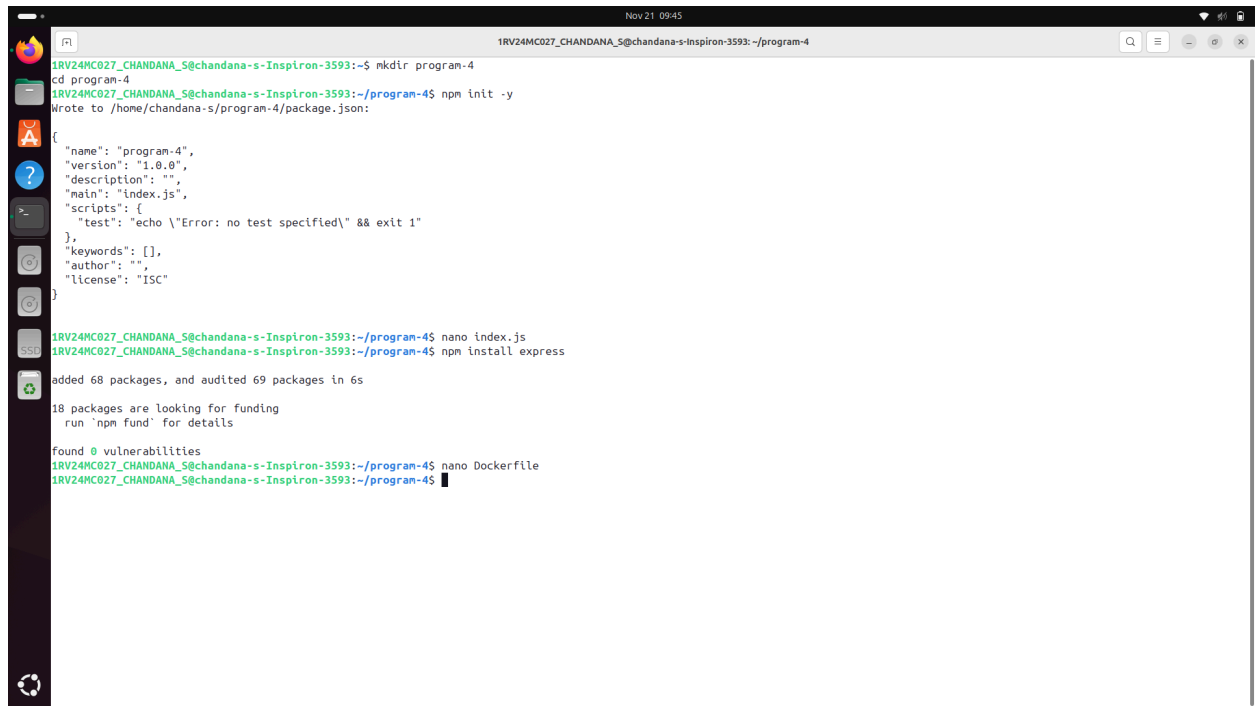
Creates a new directory named program3

`cd prog4`

Changes the working directory to program1, where all files will be created.

Step 2: Create a file package.json

`Npm init -y`



A terminal window titled "1RV24MC027_CHANDANA_S@chandana-s-Inspiron-3593: ~/program-4" showing the following commands and output:

```
1RV24MC027_CHANDANA_S@chandana-s-Inspiron-3593:~$ mkdir program-4
1RV24MC027_CHANDANA_S@chandana-s-Inspiron-3593:~/program-4$ npm init -y
Wrote to /home/chandana-s/program-4/package.json:

{
  "name": "program-4",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC"
}

1RV24MC027_CHANDANA_S@chandana-s-Inspiron-3593:~/program-4$ nano index.js
1RV24MC027_CHANDANA_S@chandana-s-Inspiron-3593:~/program-4$ npm install express
added 68 packages, and audited 69 packages in 6s

18 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
1RV24MC027_CHANDANA_S@chandana-s-Inspiron-3593:~/program-4$ nano Dockerfile
1RV24MC027_CHANDANA_S@chandana-s-Inspiron-3593:~/program-4$
```

Step 3: Create a file index.js
nano index.js

Step 4: Add Dependency
npm install express

```
1RV24MC027_CHANDANA_S@chandana-s-inspiron-3593: ~/program-4
GNU nano 7.2 index.js
const express = require("express");
const app = express();
const PORT = 3000;

app.get("/", (req, res) => {
  res.send("Hello from Git + Docker Integration!");
});

app.listen(PORT, () => {
  console.log(`Server running at http://localhost:${PORT}`);
});
```

Read 11 lines

Help Exit Write Out Read File Where Is Replace Cut Paste Execute Justify Location Go To Line Undo Redo Set Mark Copy To Bracket Where Was Previous Next Back Forward

Step 5: Create and Edit Dockerfile

nano Dockerfile



```
CNU nano 7.2
# Stage 1 - Build Stage
FROM node:18 AS build
WORKDIR /app
COPY package*.json ./
RUN npm install
COPY . .

# Stage 2 - Runtime Stage
FROM node:18-slim
WORKDIR /app
COPY --from=build /app /app
EXPOSE 3000
CMD ["node", "index.js"]
```

Step 6: Create new repo in github

Go to github.com

Click new repository

Enter name program4

Click create repository

Step 7: Getting a Personal Access Token(PAT)

GitHub → Settings → Developer settings → Personal access tokens → Tokens (classic) → Generate new token (or use

fine-grained tokens).

Give scopes repo .Copy the token — you will not be able to see it again.

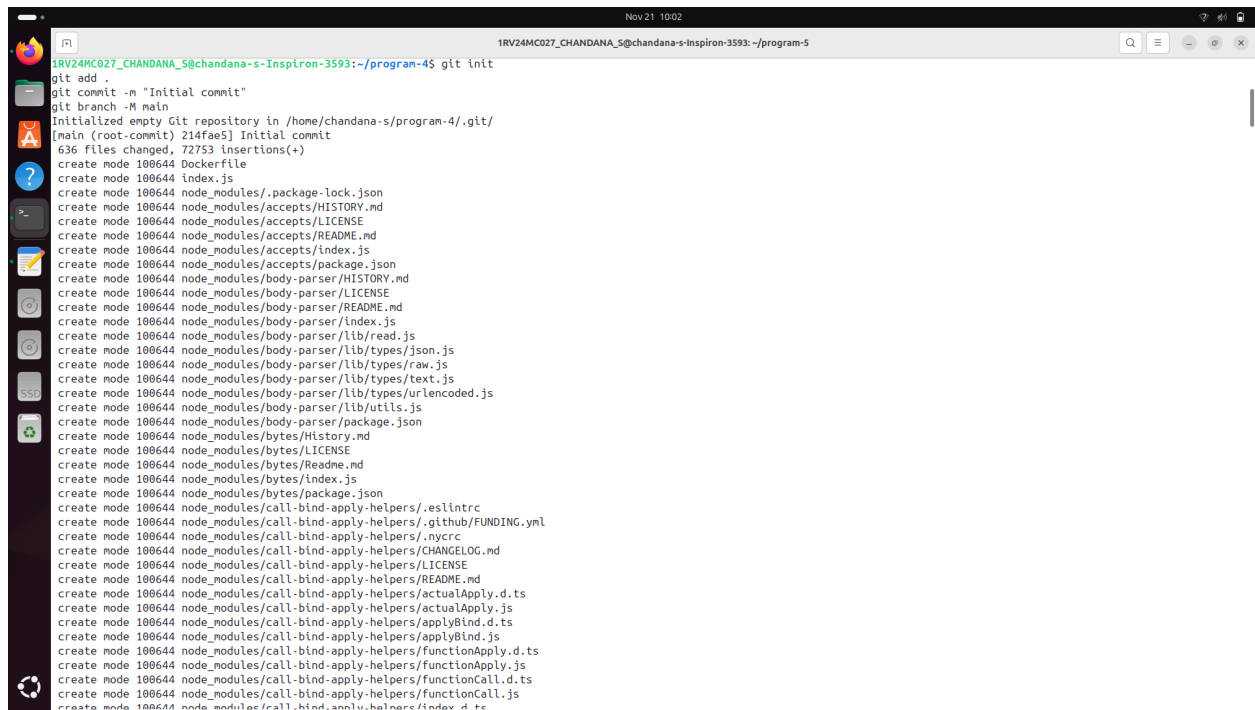
Step 8:Initialize local Git repo and commit

git init

git add .

git commit -m "Initial commit"

git branch -M main



```
1RV24MC027_CHANDANA_S@chandana-s-Inspiron-3593: ~/program-5
git init
git add .
git commit -m "Initial commit"
git branch -M main
Initialized empty Git repository in /home/chandana-s/program-4/.git/
[main (root-commit) 214fae5] Initial commit
636 files changed, 72753 insertions(+)
create mode 100644 Dockerfile
create mode 100644 index.js
create mode 100644 node_modules/.package-lock.json
create mode 100644 node_modules/accepts/HISTORY.md
create mode 100644 node_modules/accepts/LICENSE
create mode 100644 node_modules/accepts/README.md
create mode 100644 node_modules/accepts/index.js
create mode 100644 node_modules/accepts/package.json
create mode 100644 node_modules/body-parser/HISTORY.md
create mode 100644 node_modules/body-parser/LICENSE
create mode 100644 node_modules/body-parser/README.md
create mode 100644 node_modules/body-parser/index.js
create mode 100644 node_modules/body-parser/lib/read.js
create mode 100644 node_modules/body-parser/lib/types/json.js
create mode 100644 node_modules/body-parser/lib/types/raw.js
create mode 100644 node_modules/body-parser/lib/types/text.js
create mode 100644 node_modules/body-parser/lib/types/urlencoded.js
create mode 100644 node_modules/body-parser/lib/utils.js
create mode 100644 node_modules/body-parser/package.json
create mode 100644 node_modules/bytes/History.md
create mode 100644 node_modules/bytes/LICENSE
create mode 100644 node_modules/bytes/Readme.md
create mode 100644 node_modules/bytes/index.js
create mode 100644 node_modules/bytes/package.json
create mode 100644 node_modules/call-bind-apply-helpers/.eslintrc
create mode 100644 node_modules/call-bind-apply-helpers/.github/FUNDING.yml
create mode 100644 node_modules/call-bind-apply-helpers/.nycrc
create mode 100644 node_modules/call-bind-apply-helpers/CHANGELOG.md
create mode 100644 node_modules/call-bind-apply-helpers/LICENSE
create mode 100644 node_modules/call-bind-apply-helpers/README.md
create mode 100644 node_modules/call-bind-apply-helpers/actualApply.d.ts
create mode 100644 node_modules/call-bind-apply-helpers/actualApply.js
create mode 100644 node_modules/call-bind-apply-helpers/applyBind.d.ts
create mode 100644 node_modules/call-bind-apply-helpers/applyBind.js
create mode 100644 node_modules/call-bind-apply-helpers/functionApply.d.ts
create mode 100644 node_modules/call-bind-apply-helpers/functionApply.js
create mode 100644 node_modules/call-bind-apply-helpers/functionCall.d.ts
create mode 100644 node_modules/call-bind-apply-helpers/functionCall.js
create mode 100644 node_modules/call-bind-apply-helpers/index.d.ts
create mode 100644 node_modules/call-bind-apply-helpers/index.js
```

Step 9:Add Remote Repository

git remote add origin

<https://github.com/chandanas2173/program4.git>

Step 10: Push Code to GitHub

git push -u origin main

```
create mode 100644 package.json
1RV24MC027_CHANDANA_S@chandana-s-Inspiron-3593:~/program-4$ git config --global init.defaultBranch main
1RV24MC027_CHANDANA_S@chandana-s-Inspiron-3593:~/program-4$ git branch -M main
1RV24MC027_CHANDANA_S@chandana-s-Inspiron-3593:~/program-4$ git remote add origin https://github.com/chandanas2173/program4.git
1RV24MC027_CHANDANA_S@chandana-s-Inspiron-3593:~/program-4$ git push -u origin main
Username for 'https://github.com': chandanas2173
Password for 'https://chandanas2173@github.com':
Enumerating objects: 719, done.
Counting objects: 100% (719/719), done.
Delta compression using up to 8 threads
Compressing objects: 100% (666/666), done.
Writing objects: 100% (719/719), 724.81 KiB | 6.19 MiB/s, done.
Total 719 (delta 140), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (140/140), done.
remote: This repository moved. Please use the new location:
remote: https://github.com/Chandanas2173/program4.git
To https://github.com/chandanas2173/program4.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
```

